

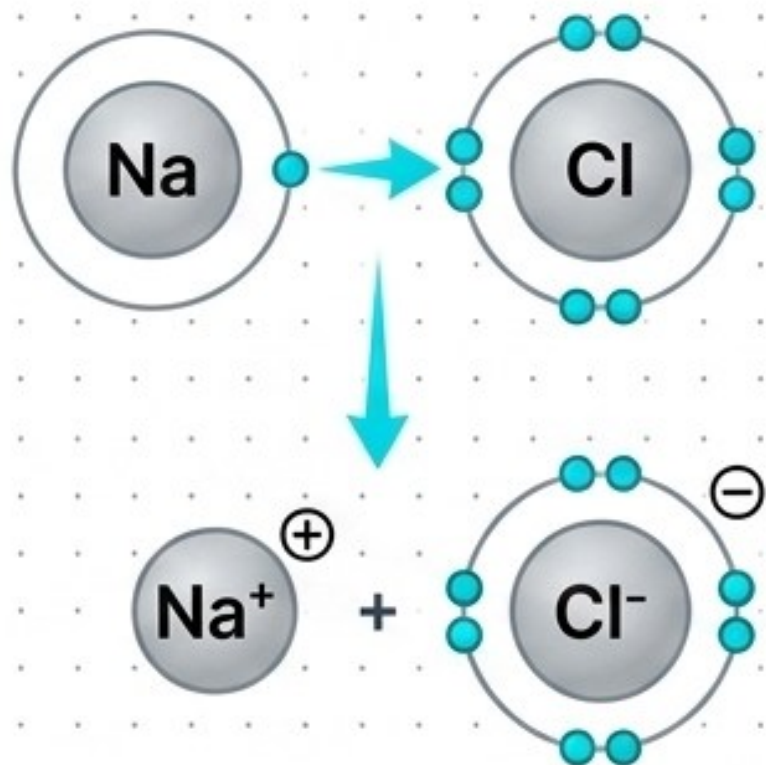
Cellular Respiration: The Metabolic Blueprint

Redox Chemistry, Energy Extraction,
and Anaerobic Pathways

Siew Ee Ling, PhD

Redox reactions power the cell by transferring electrons from organic fuels.

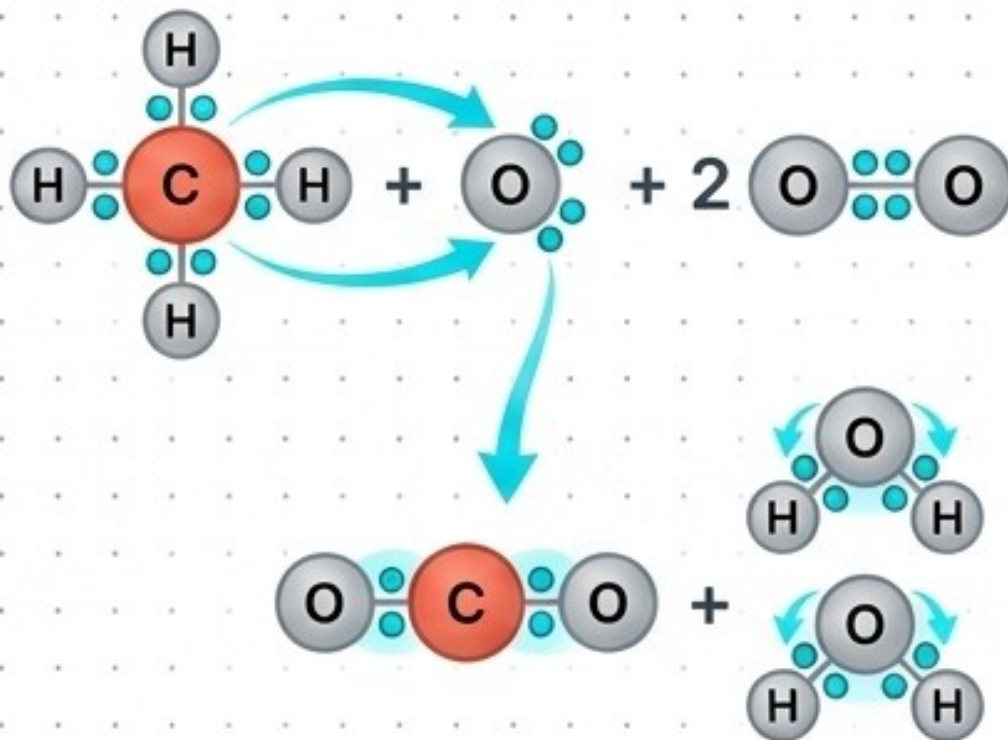
Complete Transfer



Equation: $\text{Na} + \text{Cl} \rightarrow \text{Na}^+ + \text{Cl}^-$

Mechanism: Sodium is oxidized (loses e⁻). Chlorine is reduced (gains e⁻).

Shift in Shared Electrons



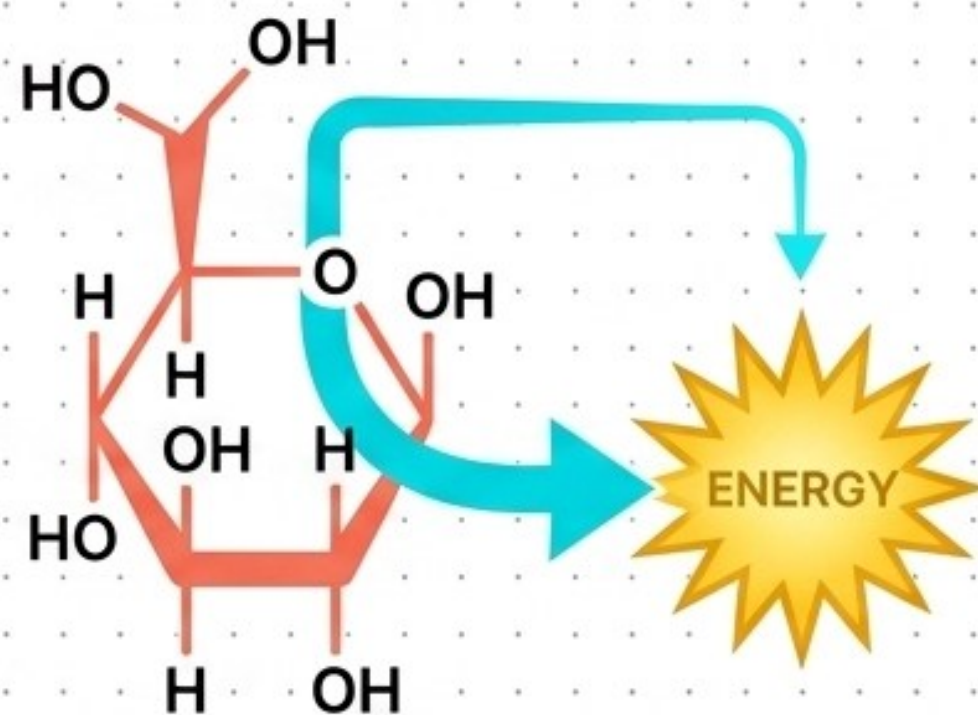
Equation: $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + \text{Energy} + 2\text{H}_2\text{O}$

Mechanism: Methane (reducing agent) is oxidized. Oxygen (oxidizing agent) is reduced. The cyan electrons physically shift closer to the electronegative oxygen atoms.

Cellular Respiration Master Equation

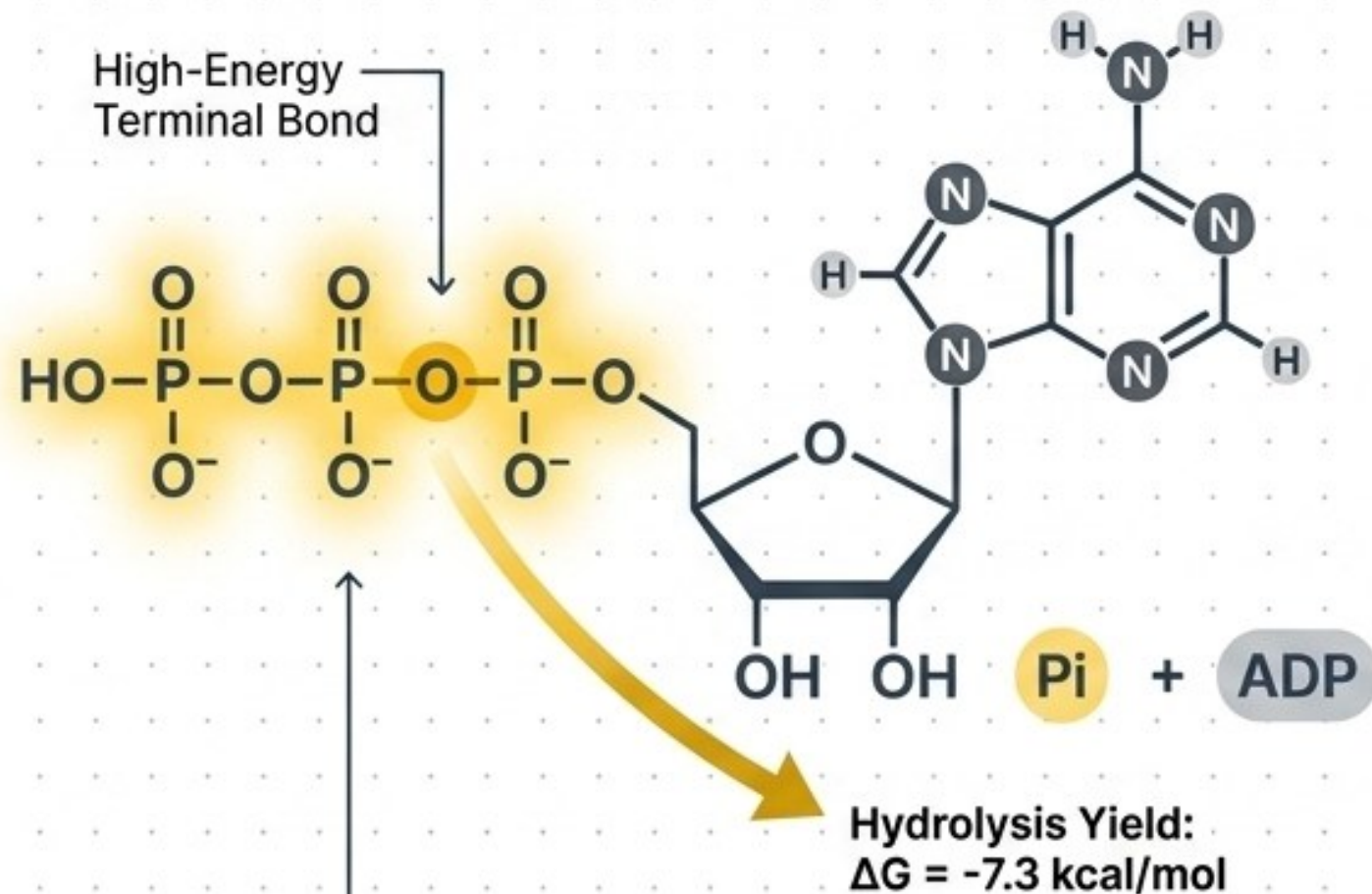
Equation: $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{Energy}$

Mechanism: Glucose is systematically oxidized. $\Delta G = -686 \text{ Kcal/mole}$.



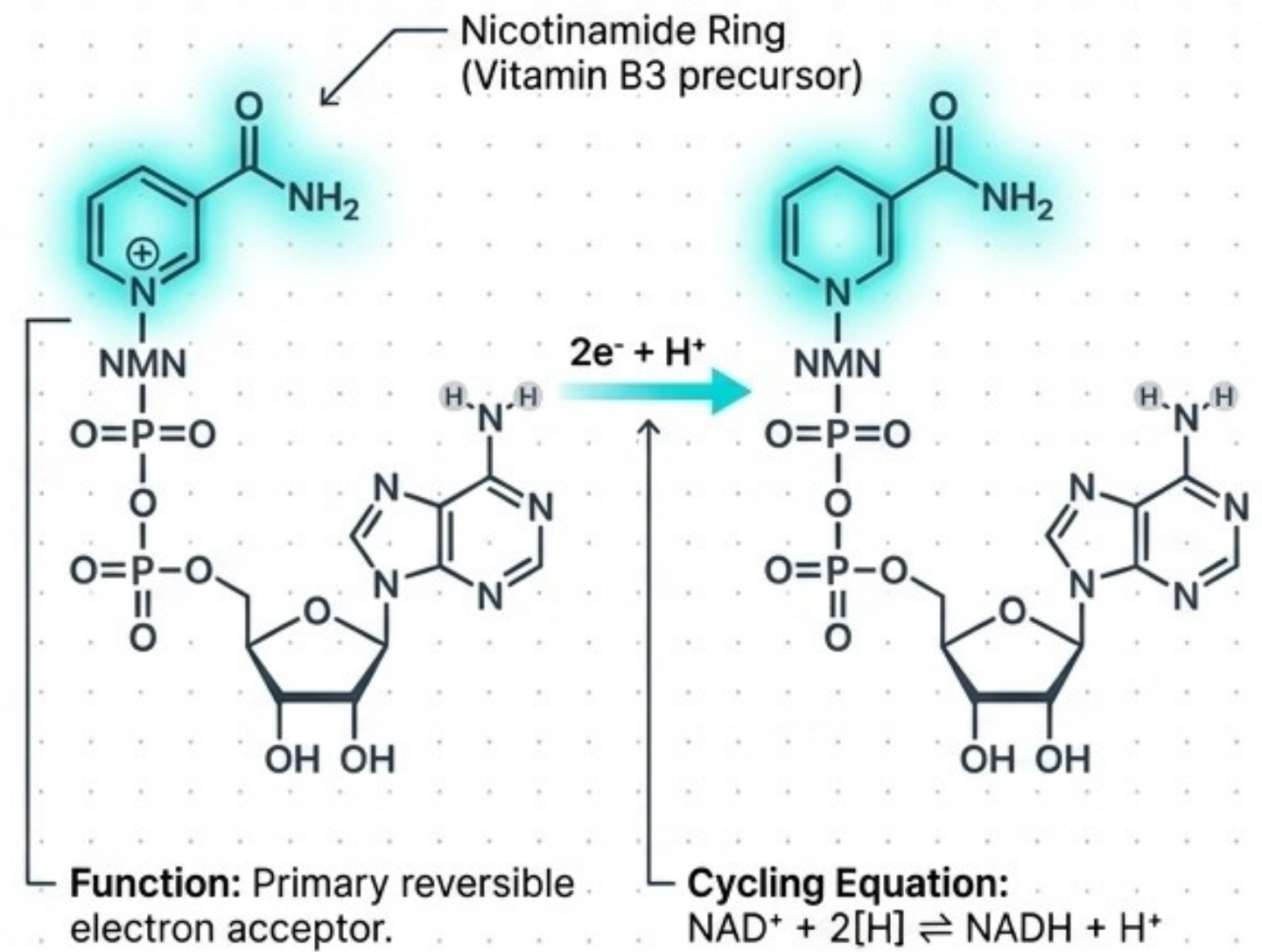
ATP provides immediate energy currency, while NAD⁺ shuttles high-energy electrons.

ATP (Adenosine Triphosphate)

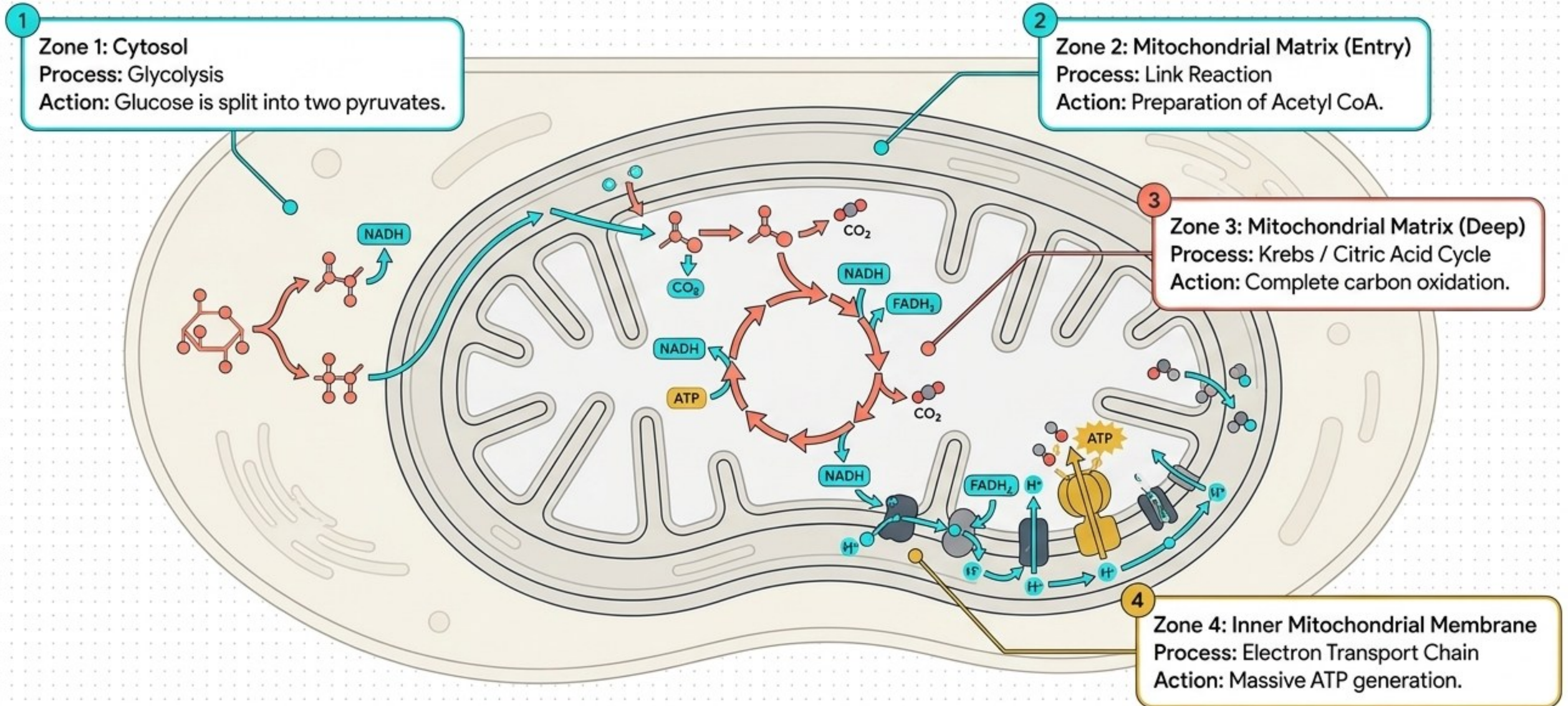


Function: Immediate energy donor for active transport and anabolic reactions.

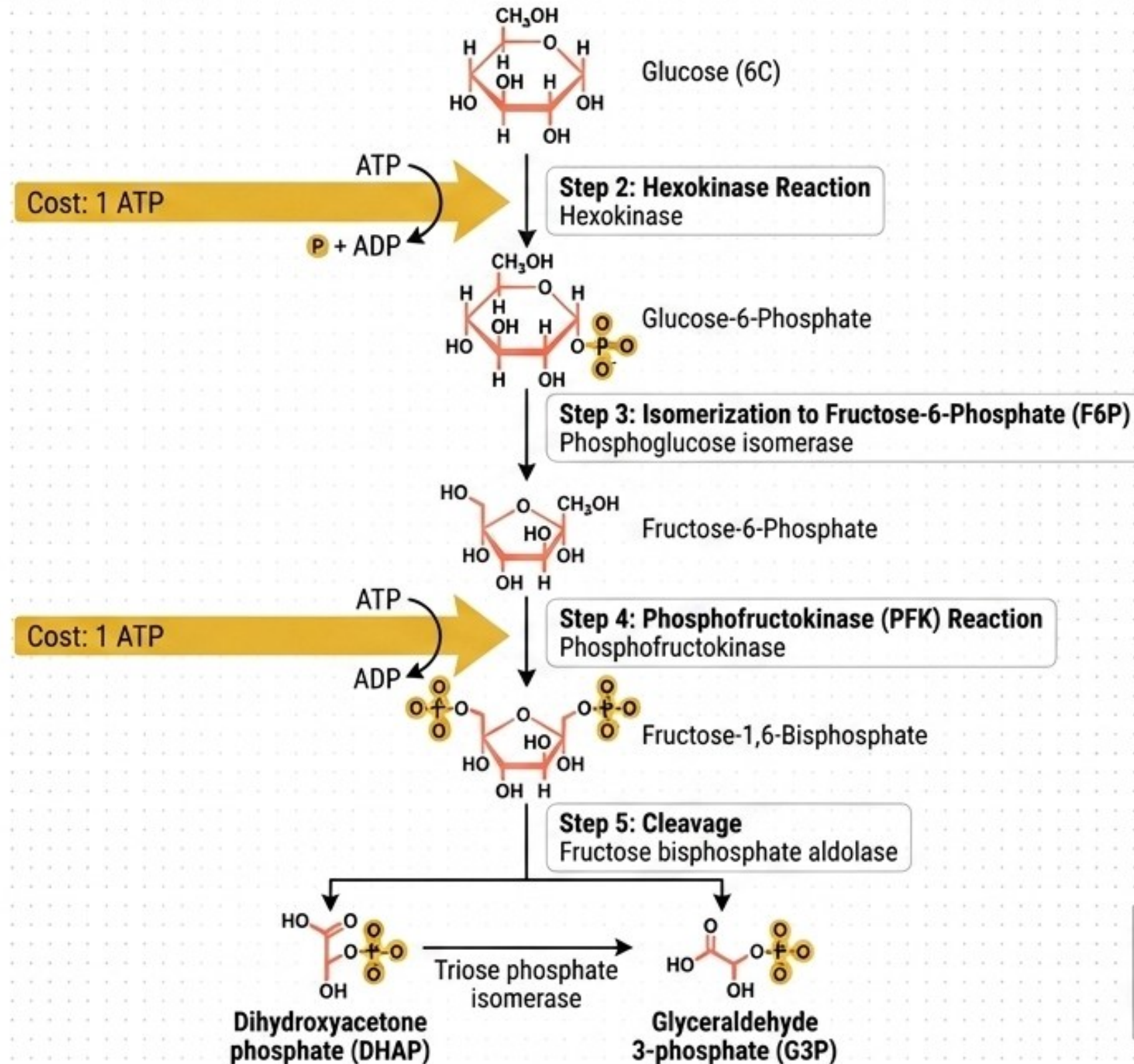
NAD⁺ (Nicotinamide Adenine Dinucleotide)



The four geographic stages of aerobic cellular respiration.

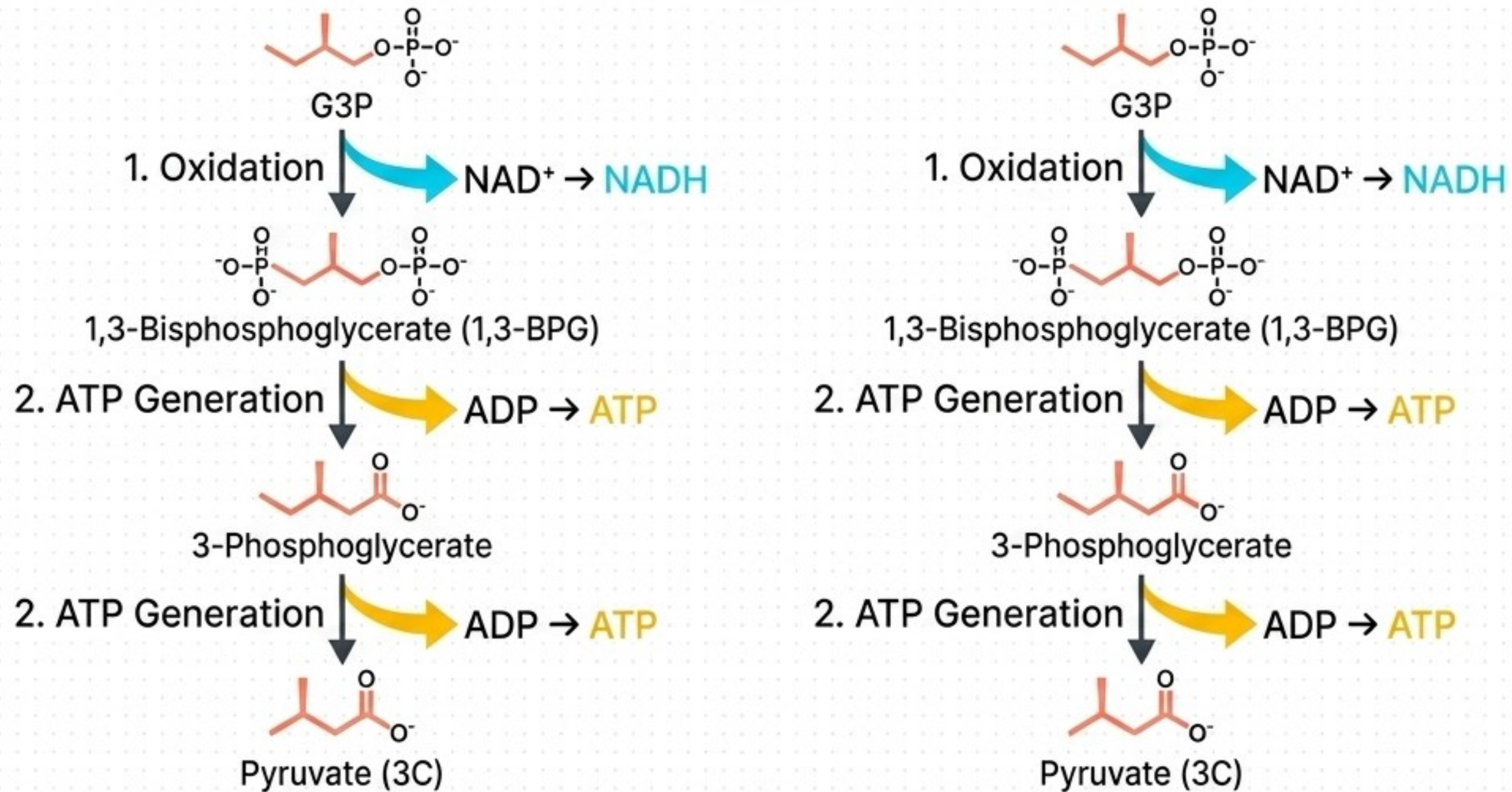


Glycolysis Phase I: The Energy Investment Phase



Key Takeaway: The cell must spend a deficit of 2 ATP to physically destabilize the glucose ring.

Glycolysis Phase II: Energy Payoff and Substrate-Level Phosphorylation



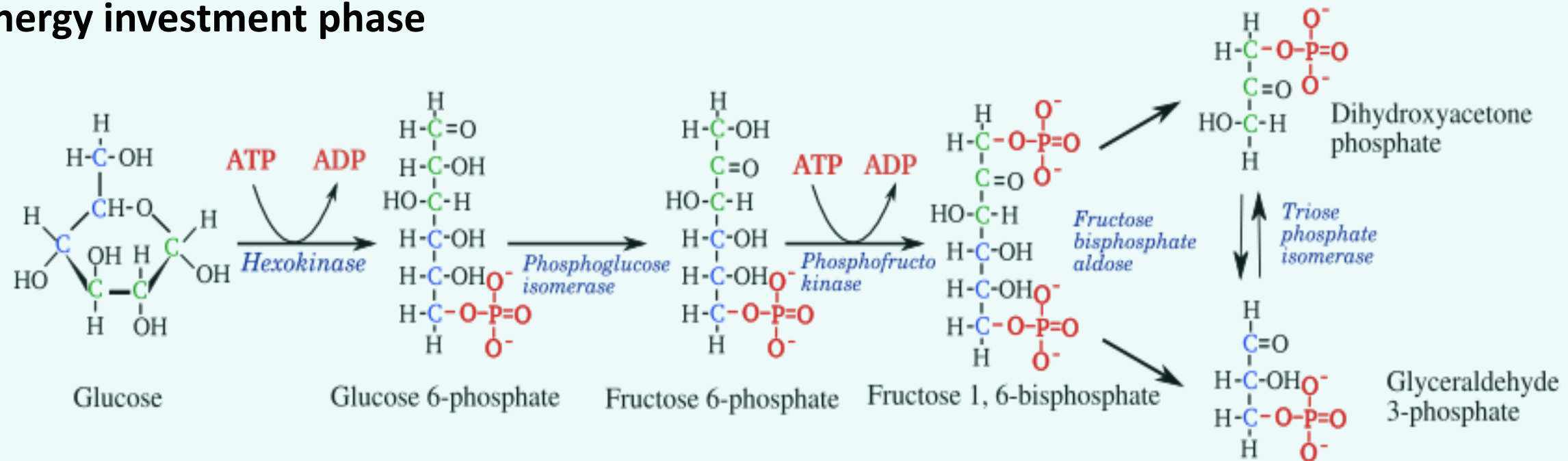
The Net Ledger (Per Glucose)

Gross Yield: **4 ATP** | **2 NADH** | 2 Pyruvate

Minus Investment: **- 2 ATP**

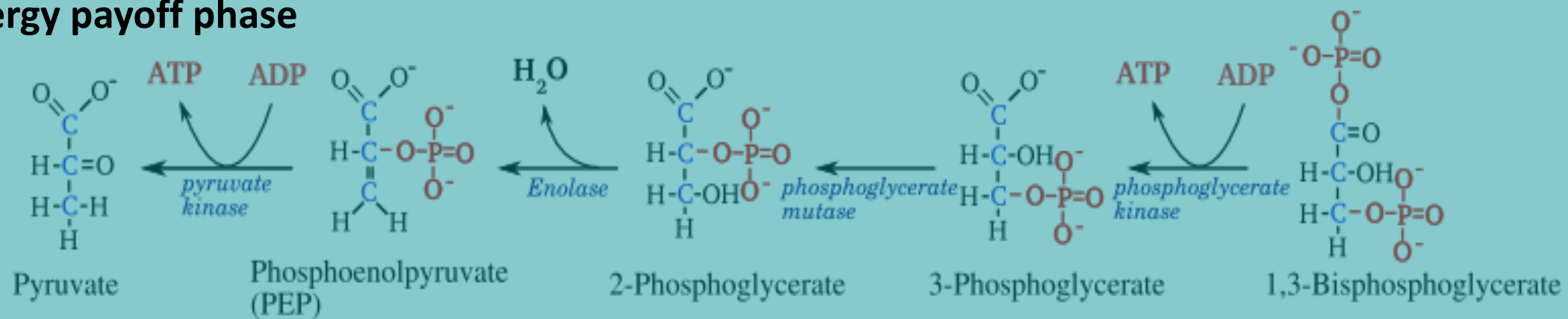
Net Yield: 2 ATP | 2 NADH

Energy investment phase

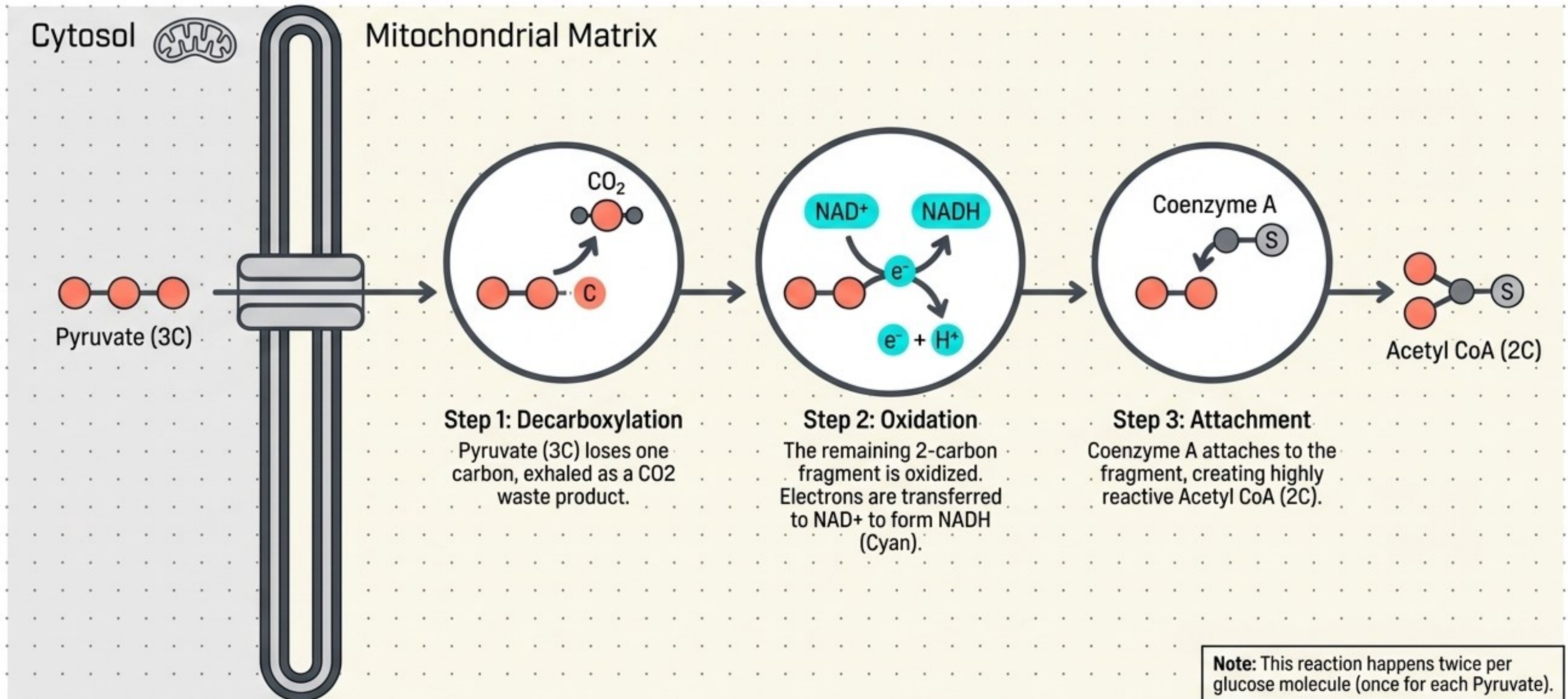


GLYCOLYSIS

Energy payoff phase



The Link Reaction: Preparing Pyruvate for the Mitochondrial Matrix







Hans Krebs who identified the citric acid cycle - also known as the Krebs cycle - was born on this day in 1900. A German biochemist. Received Noble Prize in Physiology or Medicine in 1953 for this contribution. The cycle is a series of reactions that convert nutrients into other molecules with a large amount of chemical energy.

The Citric Acid Cycle (Krebs Cycle): Complete Carbon Oxidation

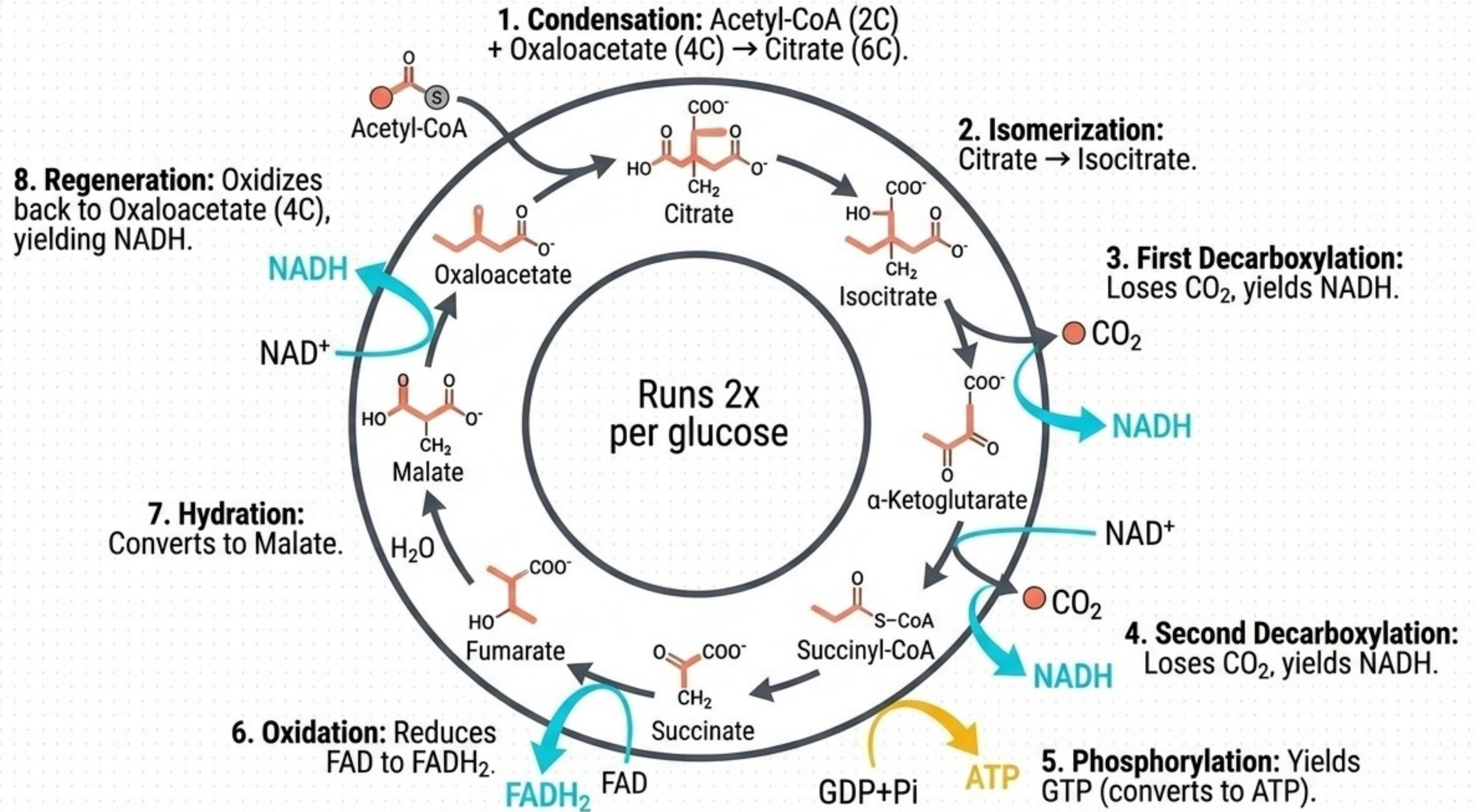


Figure 10.UN11

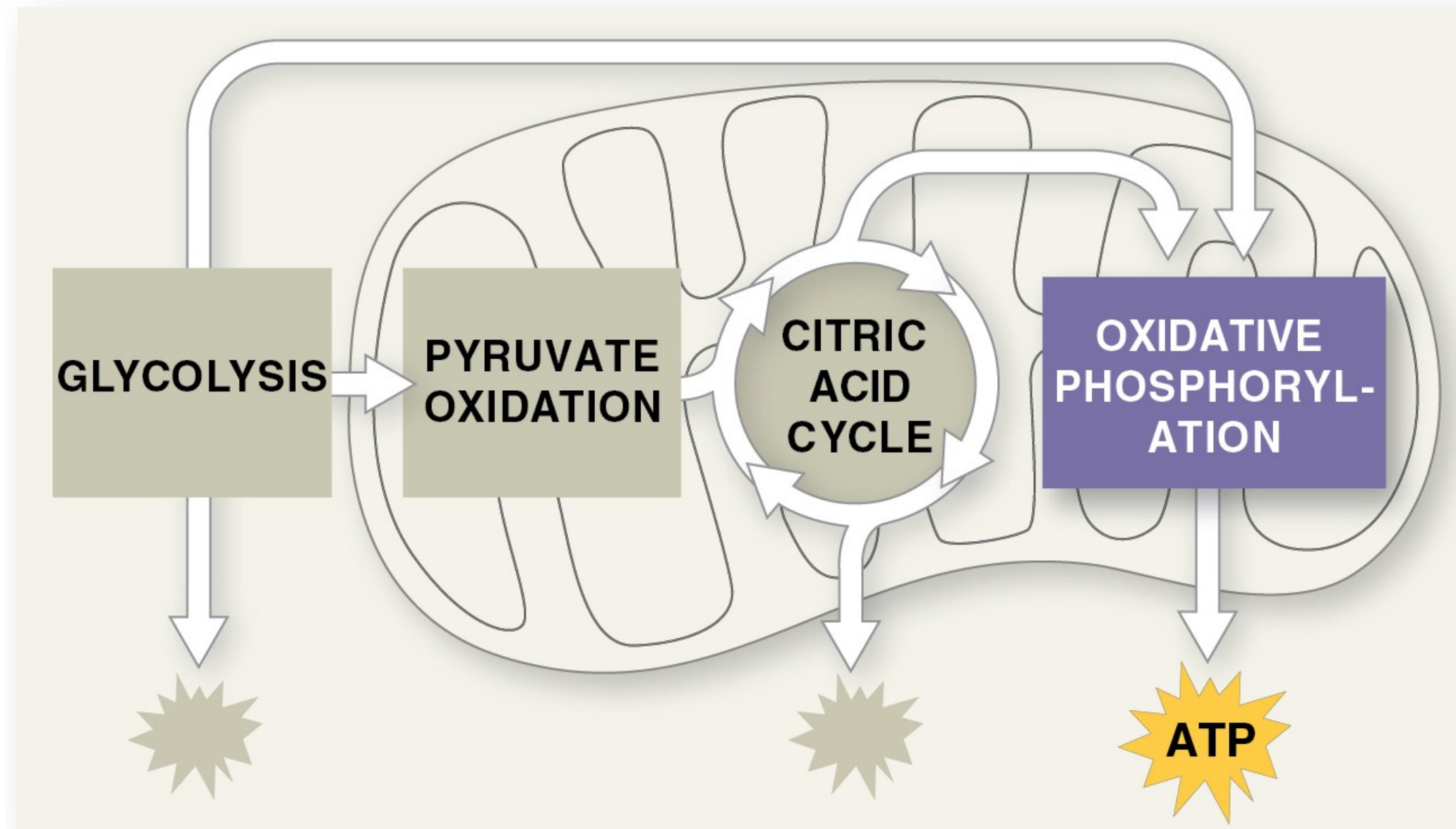
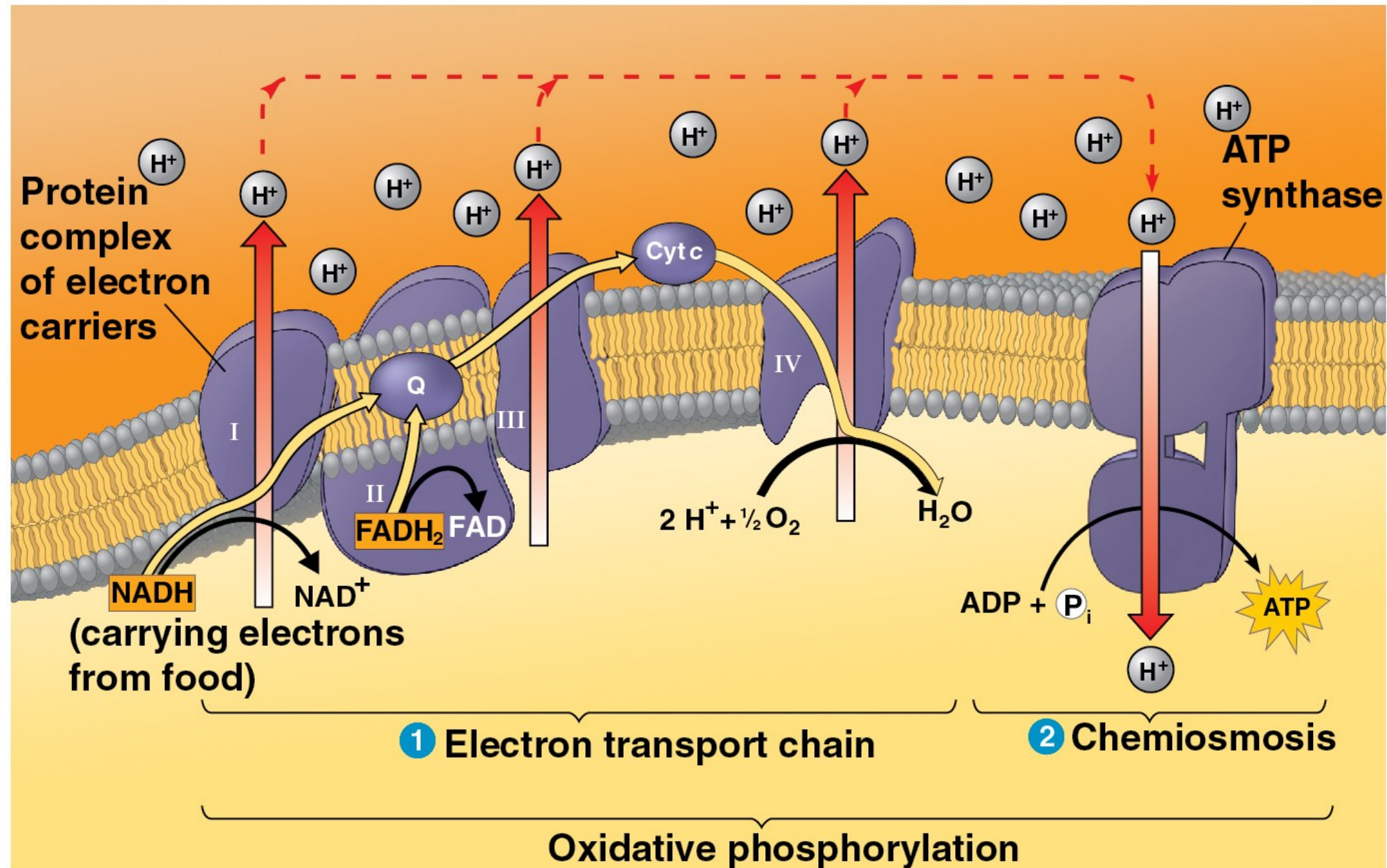
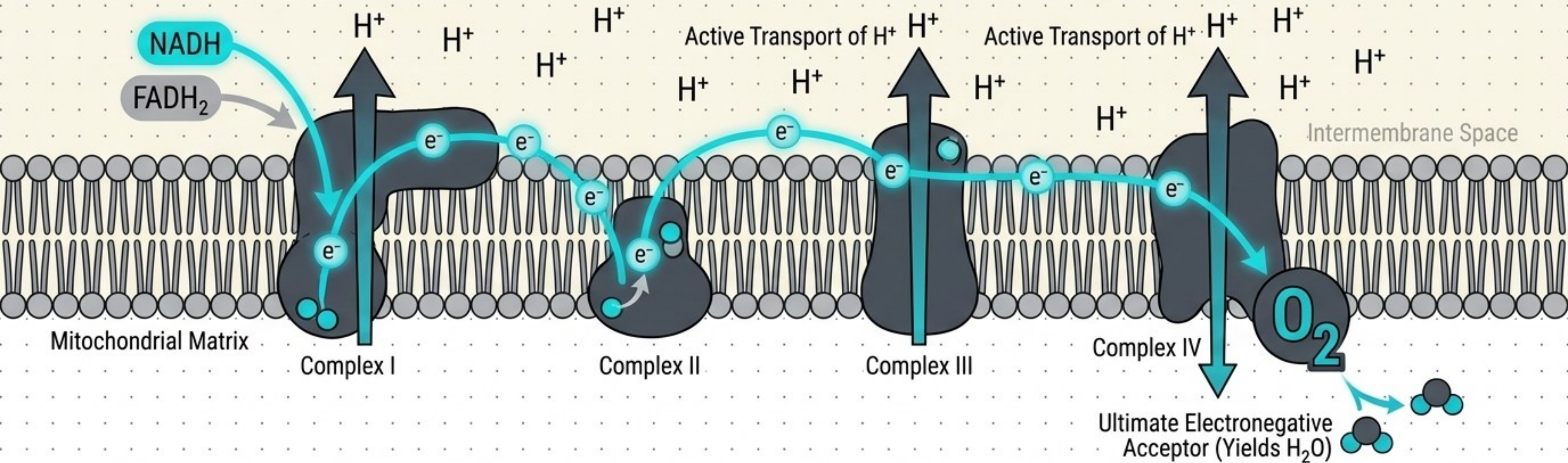


Figure 10.14



The Electron Transport Chain: Building the Proton-Motive Force

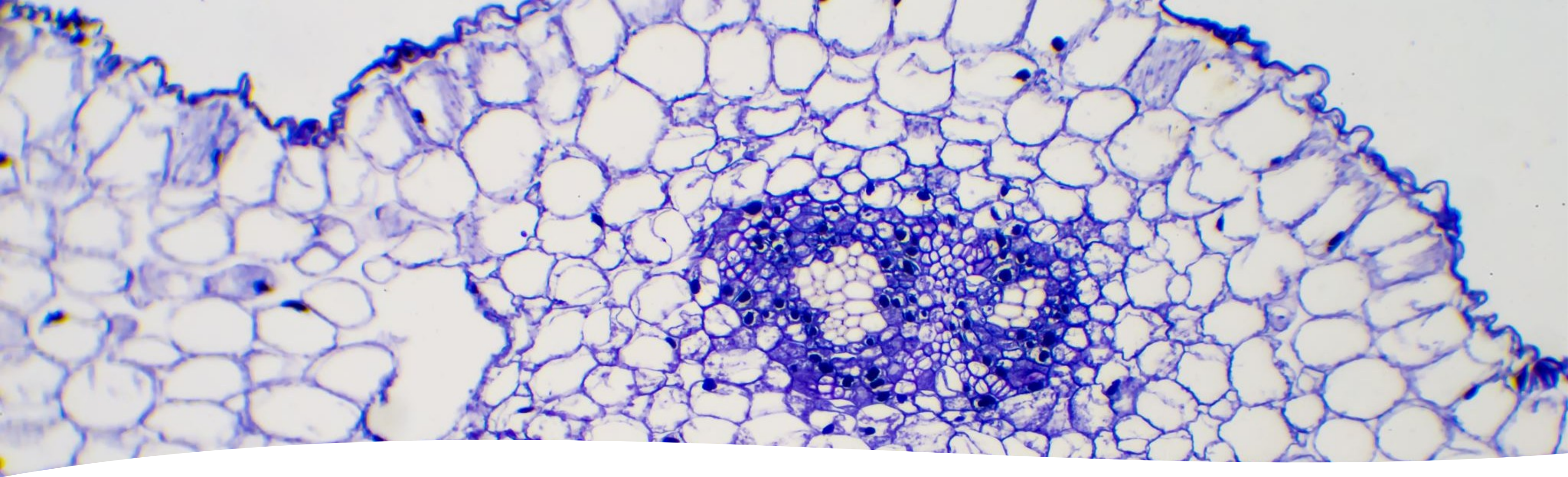


The Mechanism

1. **Electron Drop-off:** NADH and $FADH_2$ deliver electrons (cyan) to the membrane complexes.
 2. **The Electronegative Pull:** Oxygen acts as an electronegative magnet, pulling electrons down the chain.
 3. **Active Transport:** Electron energy is harnessed to pump H^+ ions against their gradient into the intermembrane space.
- Result:** An immense electrochemical H^+ gradient is formed—a pressurized battery waiting to be tapped.

ATP Production

The electrochemical gradient generated by the pumping of protons drives ATP synthesis through a process called chemiosmosis. Protons flow back into the mitochondrial matrix through ATP synthase, a protein complex that harnesses this flow to phosphorylate ADP into ATP.



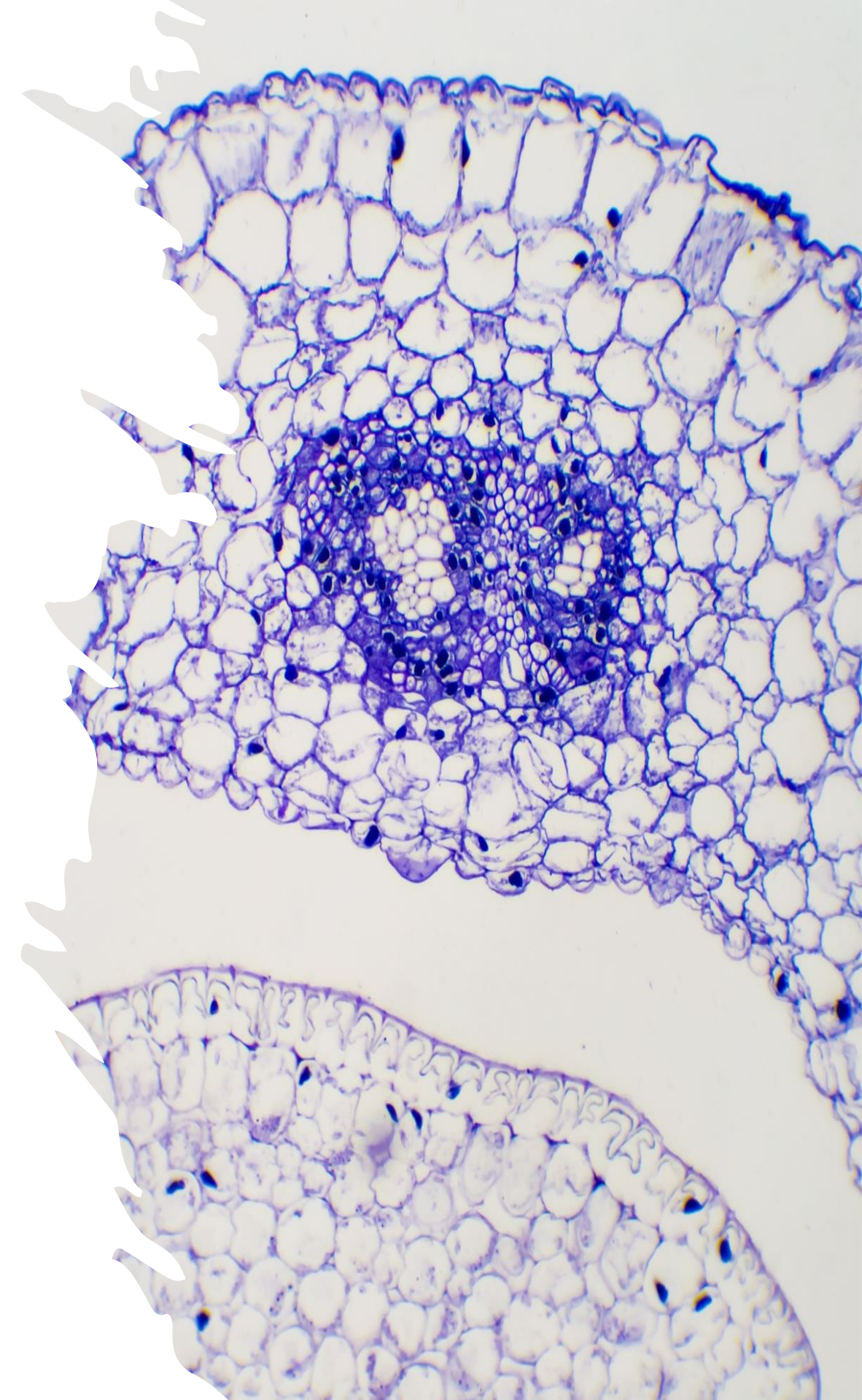
Chemiosmosis: the energy coupling mechanism

Chemiosmosis is a vital process in cellular respiration and photosynthesis, involving the movement of protons (H^+ ions) across a biological membrane, leading to the synthesis of ATP. This process is primarily associated with the electron transport chain (ETC) in mitochondria during aerobic respiration and in chloroplasts during photosynthesis.

Chemiosmosis: the energy coupling mechanism

Mechanism of Chemiosmosis

- **Proton Gradient Formation:** As electrons are transferred through the ETC, energy is released, which is used to pump protons from the mitochondrial matrix into the intermembrane space, creating a proton gradient (higher concentration of protons outside the matrix compared to inside).
- **Electrochemical Gradient:** This gradient establishes both a concentration gradient and an electrical gradient (due to the positive charge of protons), known collectively as an electrochemical gradient.
- **ATP Synthesis:** Protons naturally tend to diffuse back into the matrix due to this gradient. However, they can only do so through ATP synthase, a membrane protein that acts like a turbine. As protons flow through ATP synthase, the energy released is harnessed to convert ADP and inorganic phosphate (Pi) into ATP.



What effect would cyanide have on ATP synthesis?

- After cyanide poisoning, the electron transport chain can no longer pump electrons into the intermembrane space. The pH of the intermembrane space would increase, the pH gradient would decrease, and ATP synthesis would stop.

POISON CHEMISTRY - CYANIDE COMPOUNDS

Cyanide is a fast-acting, bitter-tasting poison, and one of the deadliest known. Its compounds have famously been utilised in suicide pills over the years.

HISTORY

Cyanide has been used for centuries as a poison, but was first identified in 1782 by the Swedish chemist Scheele; in fact, it's thought Scheele's death may have been contributed to by cyanide exposure.

During WWI, the French attempted to use hydrocyanic acid, then cyanogen chloride, in chemical warfare. In WWII, The Nazis used hydrogen cyanide in the form of Zyklon B to kill millions in their concentration camps.

Hydrogen cyanide gas has previously been used for pest control, which sometimes led to accidental deaths. Today, cyanides are still used in the mining of gold and silver, and in organic synthesis reactions.

EFFECTS

Cyanide inhibits the cytochrome oxidase enzyme, preventing the body's cells from using oxygen. HCN smells of bitter almonds, but only ~40% of people have the genetic ability to smell it. Many fruit seeds also contain small amounts of cyanide-containing compounds.

TREATMENT

The most prevalent antidotes are nitrites, which convert haemoglobin into methemoglobin, which then removes cyanide from cytochrome oxidase. Thiosulfate helps convert cyanide to thiocyanate so it can be excreted. Cobalt ions also form a stable complex with cyanide.

DETECTION

A common test for cyanide in the early 20th Century formed prussian blue, an iron-cyanide complex, in the presence of cyanide ions. There are several other chemical tests, including portable card-based tests, and instrumental methods can also be used.

HCN KCN NaCN

HYDROGEN CYANIDE POTASSIUM CYANIDE SODIUM CYANIDE
(Note: KCN and NaCN both get converted to HCN by stomach acid)

MEDIAN LETHAL DOSE: 3-8mg/kg
ORAL - VARIES DEPENDING ON FORM OF CYANIDE ADMINISTERED

AMYL NITRITE
SODIUM NITRITE
SODIUM THIOSULFATE
VITAMIN B12a

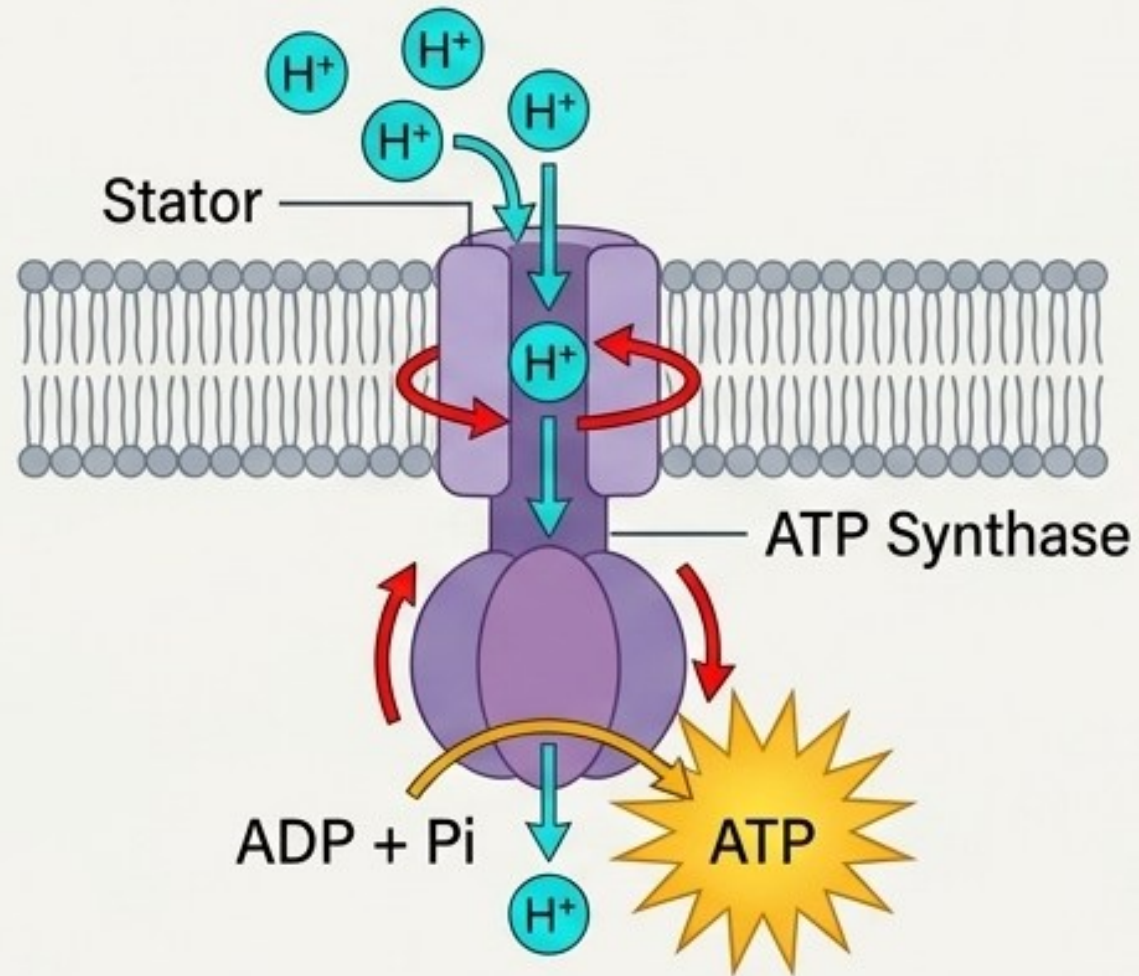
DIZZINESS & HEADACHE
NAUSEA & VOMITING
RAPID BREATHING & HEART RATE
LOW BLOOD PRESSURE
LOSS OF CONSCIOUSNESS
RESPIRATORY FAILURE
CONVULSIONS & DEATH

1 MIN
10 MINS
10-15 MINS

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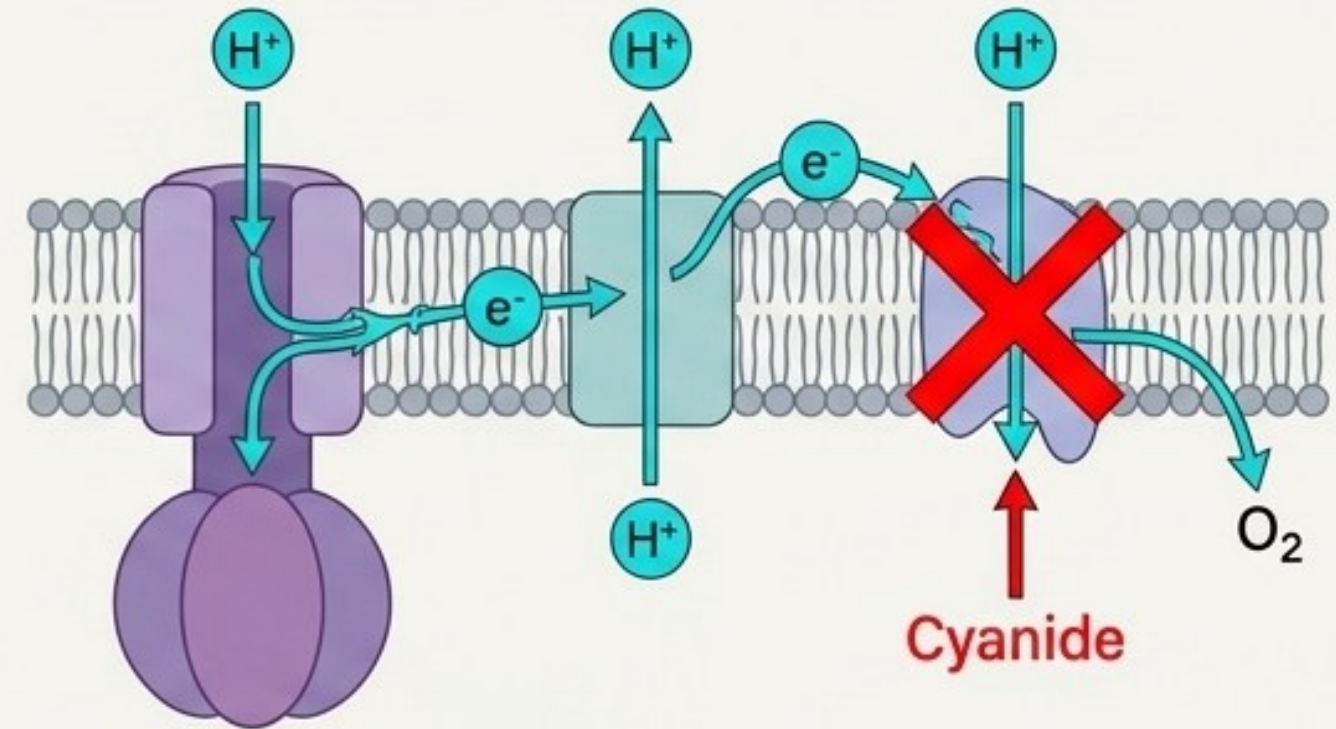
Chemiosmosis and the Cyanide Stress Test

Normal Function (Chemiosmosis)



ATP Synthase acts as a molecular rotor. H^+ ions flow down their gradient back into the matrix, spinning the stator. This physical rotation joins ADP + P_i to generate massive amounts of ATP.

Metabolic Disruption (Cyanide Poisoning)



Cyanide permanently binds to Complex IV. Electrons cannot reach Oxygen, causing the chain to back up immediately. H^+ pumping stops, the proton gradient equalizes, and ATP Synthase rotation completely halts, leading to cellular death.

An Accounting of ATP Production by Cellular Respiration

- During cellular respiration, most energy flows in this sequence:
glucose $\rightarrow$ NADH $\rightarrow$ electron transport chain $\rightarrow$ proton-motive force $\rightarrow$ ATP
- About 34% of the energy in a glucose molecule is transferred to ATP, making about 32 ATP
- The rest of the energy is lost as heat



1. Glycolysis

- **Location:** Cytoplasm
- **Net ATP Produced:** 2 ATP (from substrate-level phosphorylation)
- **NADH Produced:** 2 NADH
 - Each NADH can yield approximately 2.5 ATP in the ETC.
 - Total from NADH: $2 \times 2.5 = 5$ ATP

Total from Glycolysis:

- **ATP:** 2 (direct) + 5 (from NADH) = **7 ATP**

2. Pyruvate Oxidation

- **Location:** Mitochondrial matrix
- **NADH Produced:** 2 NADH (from the conversion of pyruvate to acetyl-CoA)
 - Total from NADH: $2 \times 2.5 = 5 \text{ ATP}$

Total from Pyruvate Oxidation:

- **ATP:** 5 ATP

3. Citric Acid Cycle (Krebs Cycle)

- **Location:** Mitochondrial matrix
- **ATP Produced:** 2 ATP (or GTP) from substrate-level phosphorylation
- **NADH Produced:** 6 NADH
 - Total from NADH: $6 \times 2.5 = 15$ ATP
- **FADH₂ Produced:** 2 FADH₂
 - Total from FADH₂: $2 \times 1.5 = 3$ ATP

Total from Citric Acid Cycle:

- **ATP:** 2 (direct) + 15 (from NADH) + 3 (from FADH₂) = **20 ATP**

Summary of Total ATP Yield

Combining the ATP yields from all stages:

- Glycolysis: 7 ATP
- Pyruvate Oxidation: 5 ATP
- Citric Acid Cycle: 20 ATP

Total ATP Yield from One Glucose Molecule:

- $7 + 5 + 20 = 32 \text{ ATP}$

The "Exchange Rate" Concept

Direct Cash vs. Checks

ATP acts as the cell's direct cash currency used immediately for biological work.

NADH and **FADH₂** are high-energy electron carriers (like uncashed checks). They cannot be spent directly and must be processed through the Electron Transport Chain (ETC).

Indirect Coupling Mechanism

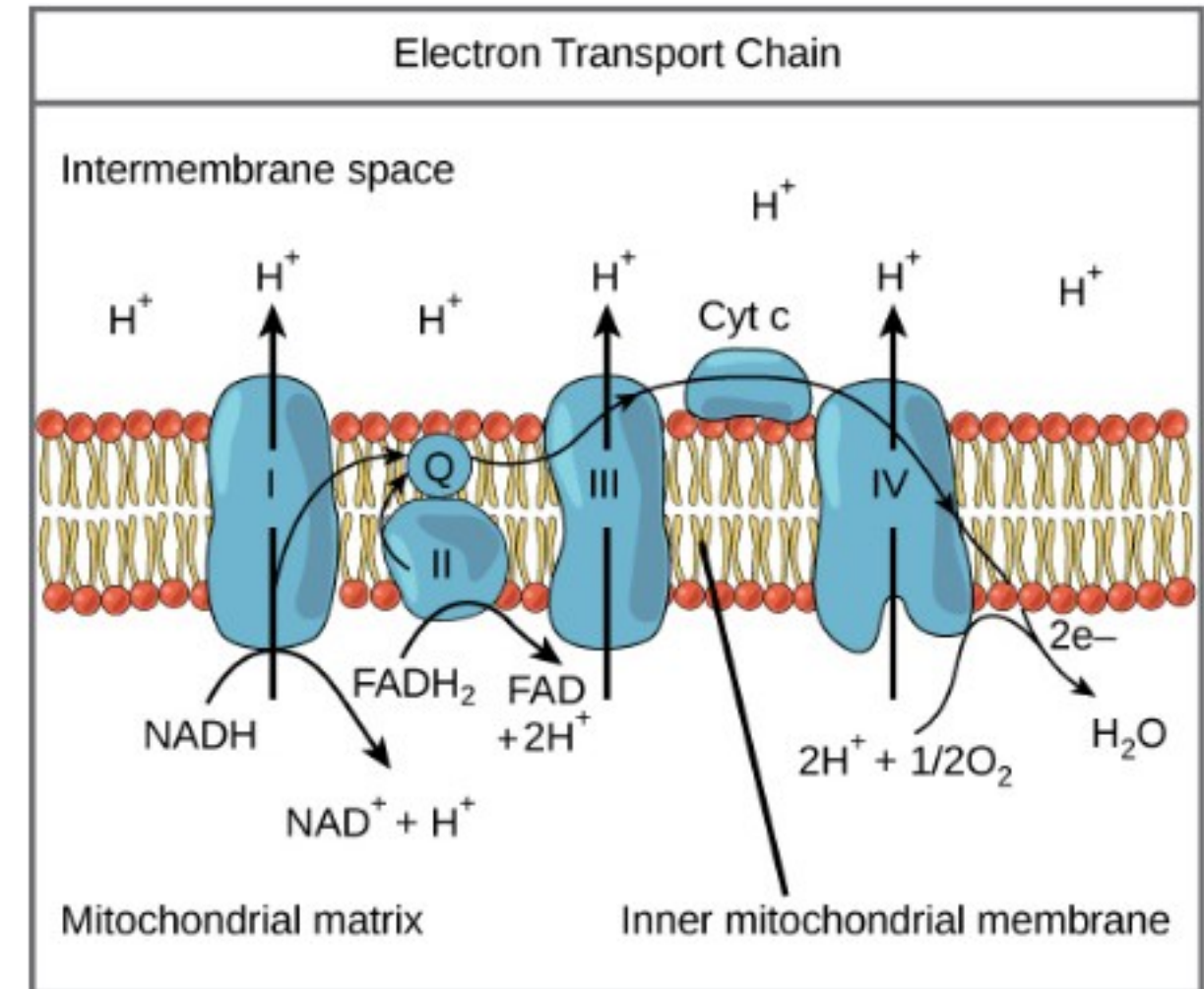
Cellular respiration uses a **proton pump gradient** (chemiosmosis) rather than a 1:1 chemical reaction.

Because ATP is synthesized indirectly as protons flow through **ATP Synthase**, the conversion factors result in fractional stoichiometry rather than round integers.

ETC Entry Points Breakdown

Why FADH₂ Yields Less Energy

- **NADH enters at Complex I:** Donates high-energy electrons early, triggering proton pumping across Complex I (4 H⁺), Complex III (4 H⁺), and Complex IV (2 H⁺), totaling **10 H⁺**.
- **FADH₂ enters at Complex II:** Bypasses Complex I entirely. Complex II does not pump any protons.
- **Reduced Proton Count:** Electrons from FADH₂ only power Complex III (4 H⁺) and Complex IV (2 H⁺), yielding a total of **6 H⁺**.



The Proton Pumping Math

Electron Carrier	Path in ETC	Protons Pumped (H+)	H+ Required per ATP	ATP Conversion Rate
NADH	Complexes I, III, IV	10 H+	4 H+	2.5 ATP
FADH2	Complexes II, III, IV	6 H+	4 H+	1.5 ATP

ATP Yield per Carrier = $\frac{\text{Total Protons Pumped (H}^+\text{)}}{\text{Protons Required per ATP (4H}^+\text{)}}$

*Note: Synthesis of 1 ATP requires 3 H+ passing through ATP Synthase plus 1 H+ used for phosphate/ADP transport into the matrix.

NADH yields 2.5 ATP; FADH₂ yields 1.5 ATP

ATP is the cell’s spendable energy currency. NADH and FADH₂ carry high-energy electrons that must be “cashed in” through the electron transport chain.

ENTRY POINT	PROTON-PUMPING COMPLEXES	TOTAL H ⁺ PUMPED
NADH	Complex I → Complex III → Complex IV	10 H ⁺
FADH ₂	Complex II (no H ⁺ pumped) → Complex III → Complex IV	6 H ⁺

NADH enters at Complex I

$10\text{ H}^+ \div 4\text{ H}^+ \text{ per ATP} = 2.5\text{ ATP}$

Electrons pass through Complexes I, III and IV.
All three complexes pump protons across the inner membrane.
≈4 H⁺ are required per ATP, including transport costs.

FADH₂ enters at Complex II

$6\text{ H}^+ \div 4\text{ H}^+ \text{ per ATP} = 1.5\text{ ATP}$

Complex II transfers electrons but does not pump protons.
Electrons therefore drive pumping only at Complexes III and IV.
Fewer pumped protons produce less ATP per electron carrier.

Modern P/O ratios: 2.5 ATP per NADH and 1.5 ATP per FADH₂ account for proton stoichiometry and transport/coupling costs. Older texts often use 3 and 2 ATP.

The Energy Ledger: Accounting for Maximum ATP Yield

Stage	Location	Electron Carriers (Cyan)	Direct ATP (Gold)
Glycolysis	Cytosol	2 NADH	2 ATP
Link Reaction	Matrix	2 NADH	0 ATP
Citric Acid Cycle	Matrix	6 NADH, 2 FADH ₂	2 ATP
Oxidative Phosphorylation (Inner Membrane): 10 NADH and 2 FADH ₂ cash out at the ETC, generating ~26-28 ATP.			

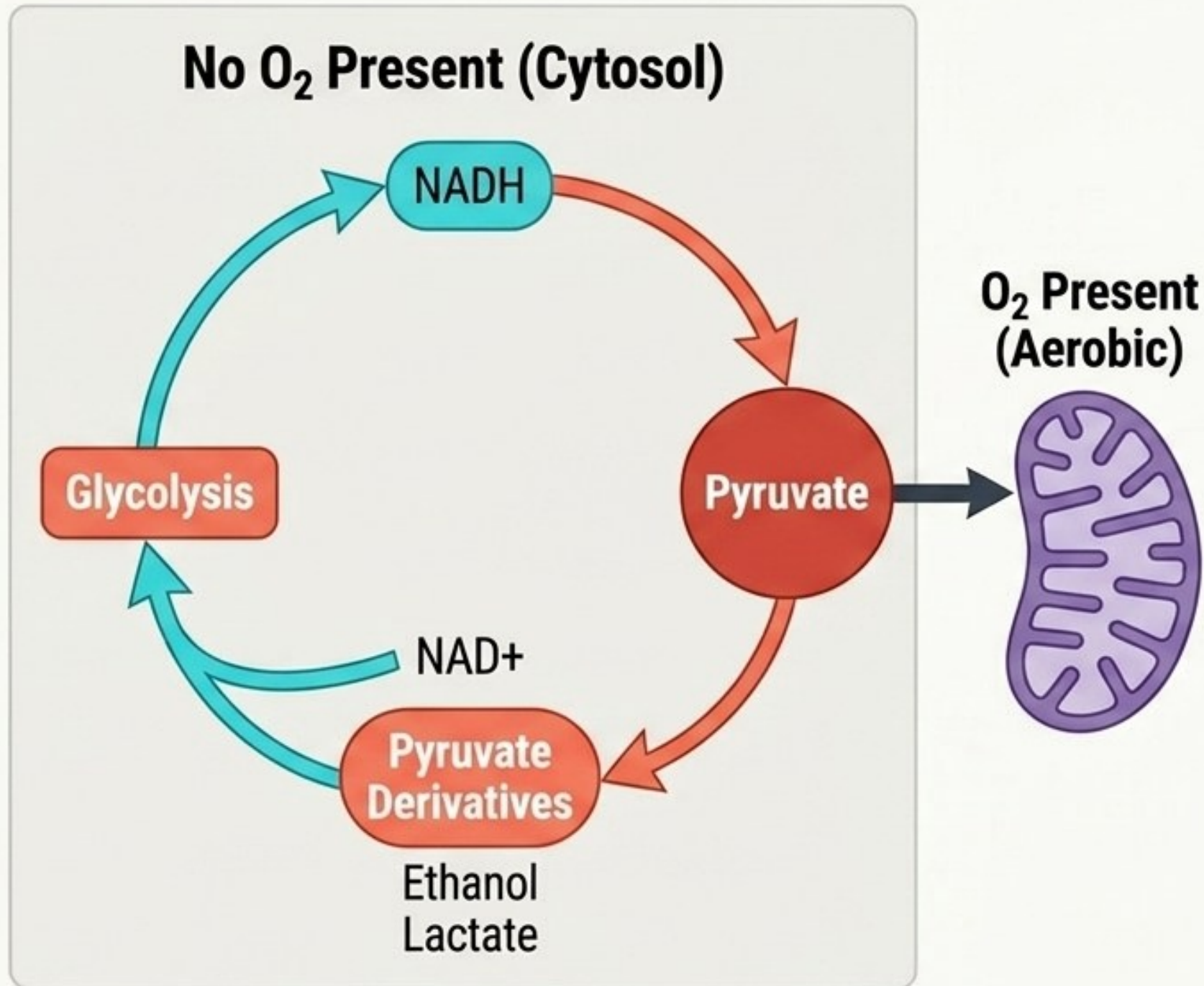
Total Output: 30-32 ATP per single molecule of glucose.

Efficiency Note: Oxidative phosphorylation accounts for the vast majority of extracted energy; without electronegative oxygen, the chain ceases to operate.

Where does the figure of 30-32 ATP come from?

- Two net ATP are made in **glycolysis**, and another two ATP (or energetically equivalent GTP) are made in the **citric acid cycle**.
- Beyond those four, the remaining ATP all come from **oxidative phosphorylation**. Based on a lot of experimental work, it appears that four H⁺ ions must flow back into the matrix through ATP synthase to power the synthesis of one ATP molecule. When electrons from NADH move through the transport chain, about 10 H⁺ ions are pumped from the matrix to the intermembrane space, so each NADH yields about **2.5 ATP**. Electrons from FADH₂ which enter the chain at a later stage, drive pumping of only 6 H⁺ leading to production of about **1.5 ATP**.

Anaerobic Respiration & Fermentation: Surviving Without Oxygen



The Primary Problem

- Without Oxygen, the ETC stops. NADH backs up, leaving no NAD⁺ to accept electrons in Glycolysis. ATP production flatlines.

The Fermentation Solution

- Fermentation is an extension of glycolysis designed to oxidize NADH by transferring its electrons back to pyruvate derivatives.
- **Goal:** Regenerate NAD⁺ so glycolysis can continue producing its net 2 ATP.

True Anaerobic Respiration

- Distinct from fermentation, some obligate anaerobes use an ETC but utilize a different final electron acceptor (e.g., Sulfate ion SO₄²⁻, yielding H₂S instead of H₂O).

Comparison Matrix: Fermentation Pathways

Dimension	Alcohol Fermentation	Lactic Acid Fermentation
Pathway	Pyruvate → Acetaldehyde → Ethanol	Pyruvate directly reduced to Lactate
Byproducts	Releases CO2 (causes bread to rise)	No CO2 released
Organisms / Context	Yeast and some bacteria	Fungi, certain bacteria, human muscle cells under oxygen debt
Net ATP Yield	2 ATP (derived purely from the glycolysis cycle)	2 ATP (derived purely from the glycolysis cycle)

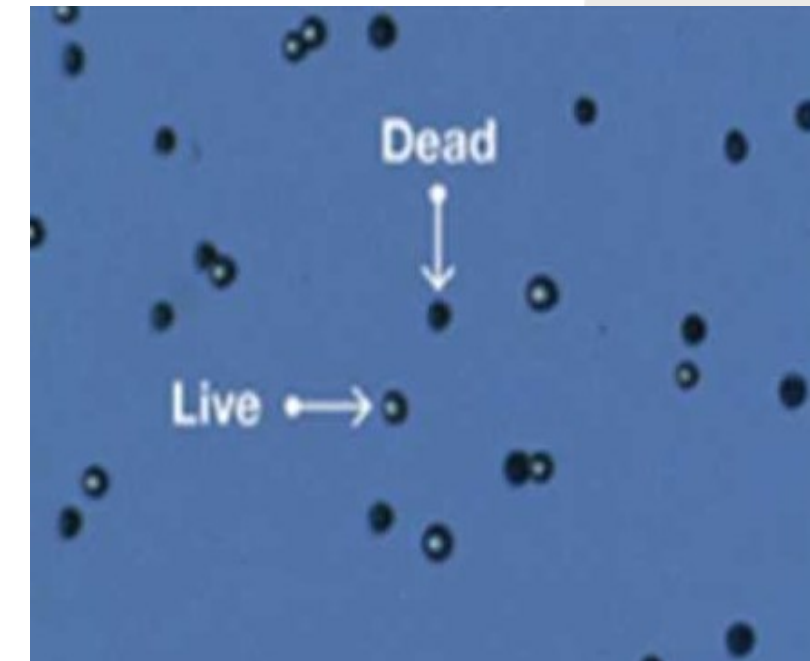
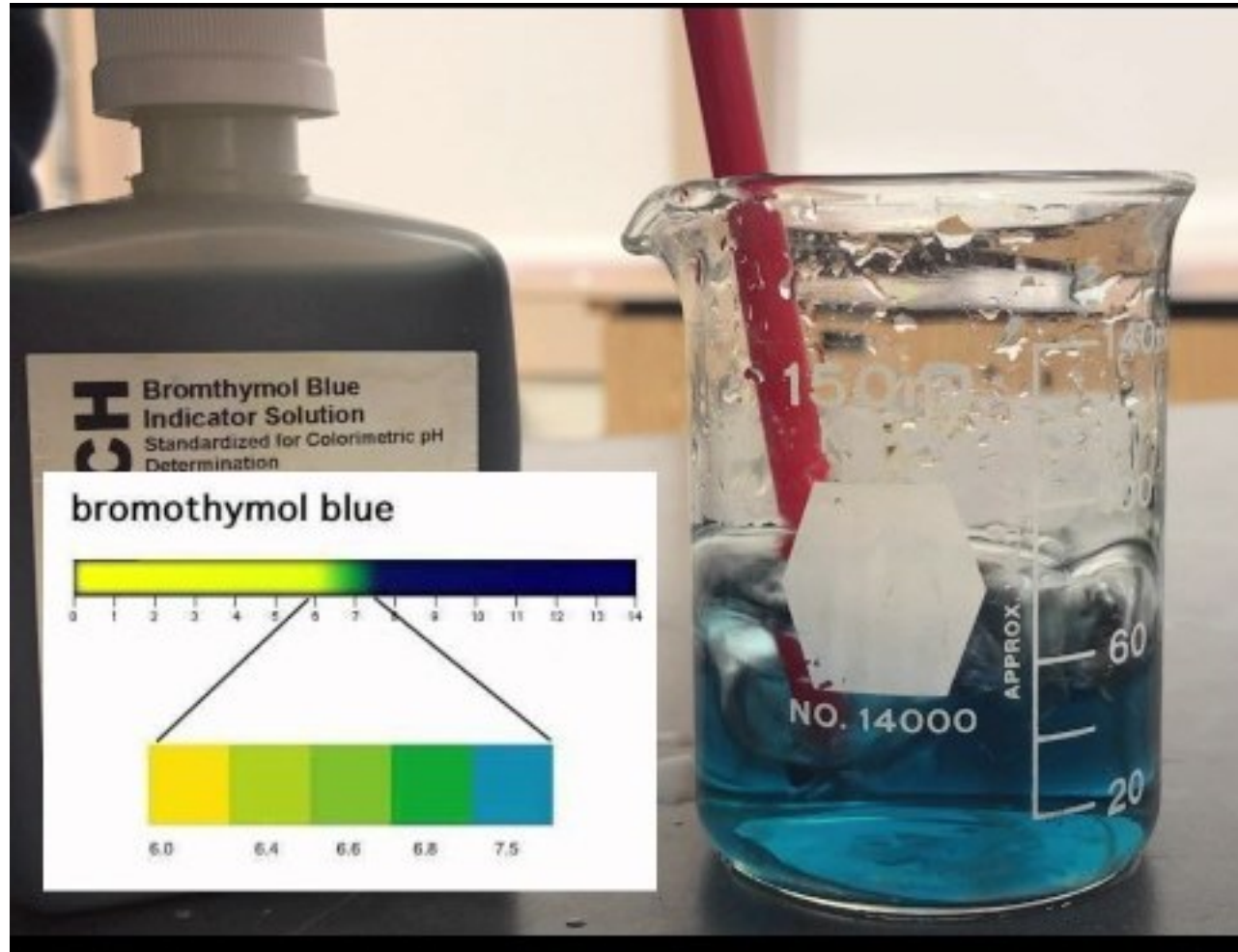
FUN FACT (STUDENT DEMO)

Bromophenol blue is that it not only serves as a pH indicator but also has a unique application in biology: it can be used to stain dead cells blue! This property is particularly useful in microscopy for assessing cell viability.

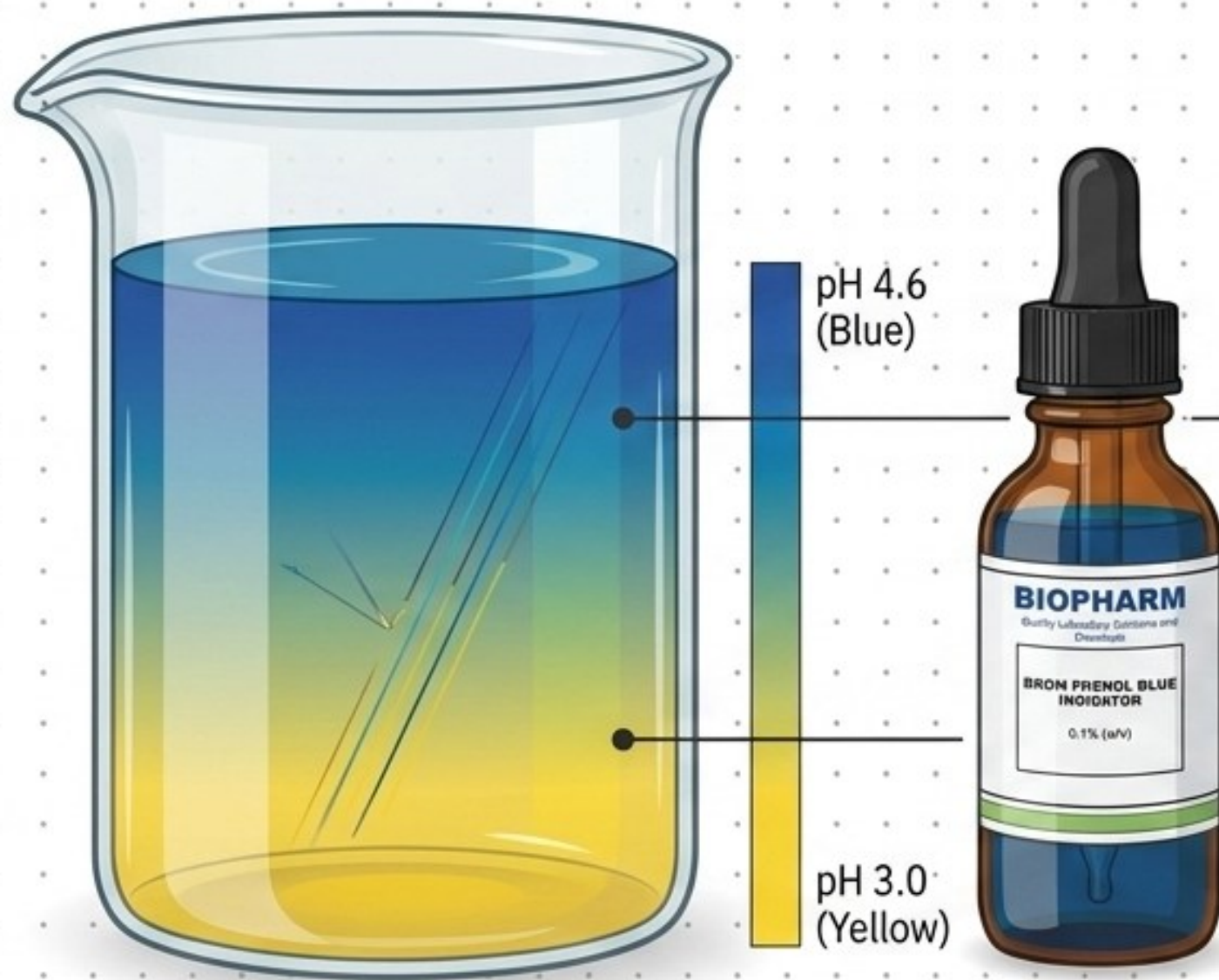
During cellular respiration, organisms convert glucose and oxygen into energy, producing carbon dioxide and water as byproducts. The increase in CO₂ concentration in a closed system (like a solution containing bromophenol blue) results in:

Formation of Carbonic Acid: CO₂ dissolves in water, forming carbonic acid, which decreases the pH.

Color Change: As the pH drops, bromophenol blue shifts from blue to yellow, indicating increased acidity.



Lab Application: Visualizing Respiration with Bromophenol Blue



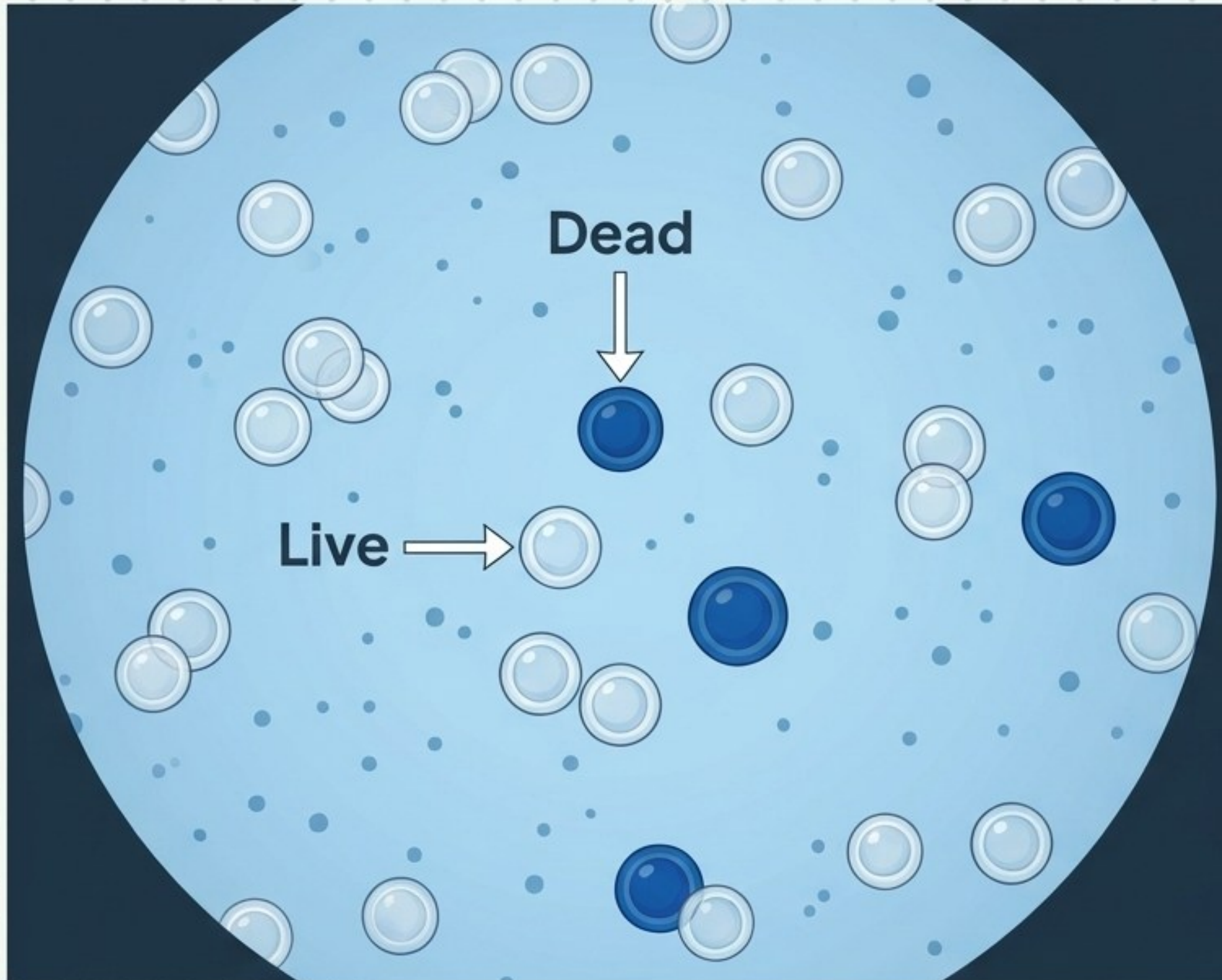
• The Chemical Mechanism

1. Organisms consume O_2 and exhale CO_2 into a closed aqueous system.
2. CO_2 dissolves in water to form Carbonic Acid (H_2CO_3) (**Carbon**: rich coral).
3. The pH of the solution steadily decreases.

• The Indicator Shift

- Bromophenol blue tracks this acidification in real-time.
- Starts **Blue** at **pH 4.6** (low respiration / low CO_2) (Electron carriers: vibrant cyan).
- Shifts to **Yellow** at **pH 3.0** (high respiration / carbonic acid buildup) (ATP and energy: bright golden-yellow).

Lab Application: Dual-Purpose Viability Staining



The Biological Mechanism

Beyond acting as a pH indicator, Bromophenol blue distinguishes live from dead cells in microscopy.

Live Cells (Translucent)

Possess intact, semi-permeable membranes and active ATP-powered pumps that continuously exclude the dye from the cytoplasm.

Dead Cells (Stained Blue)

Suffer from compromised, porous lipid membranes. Without ATP, active pumps fail, allowing the dye to passively flood the cell.

Conclusion

The presence of functional cellular respiration literally keeps the dye at bay.

Take home messages!

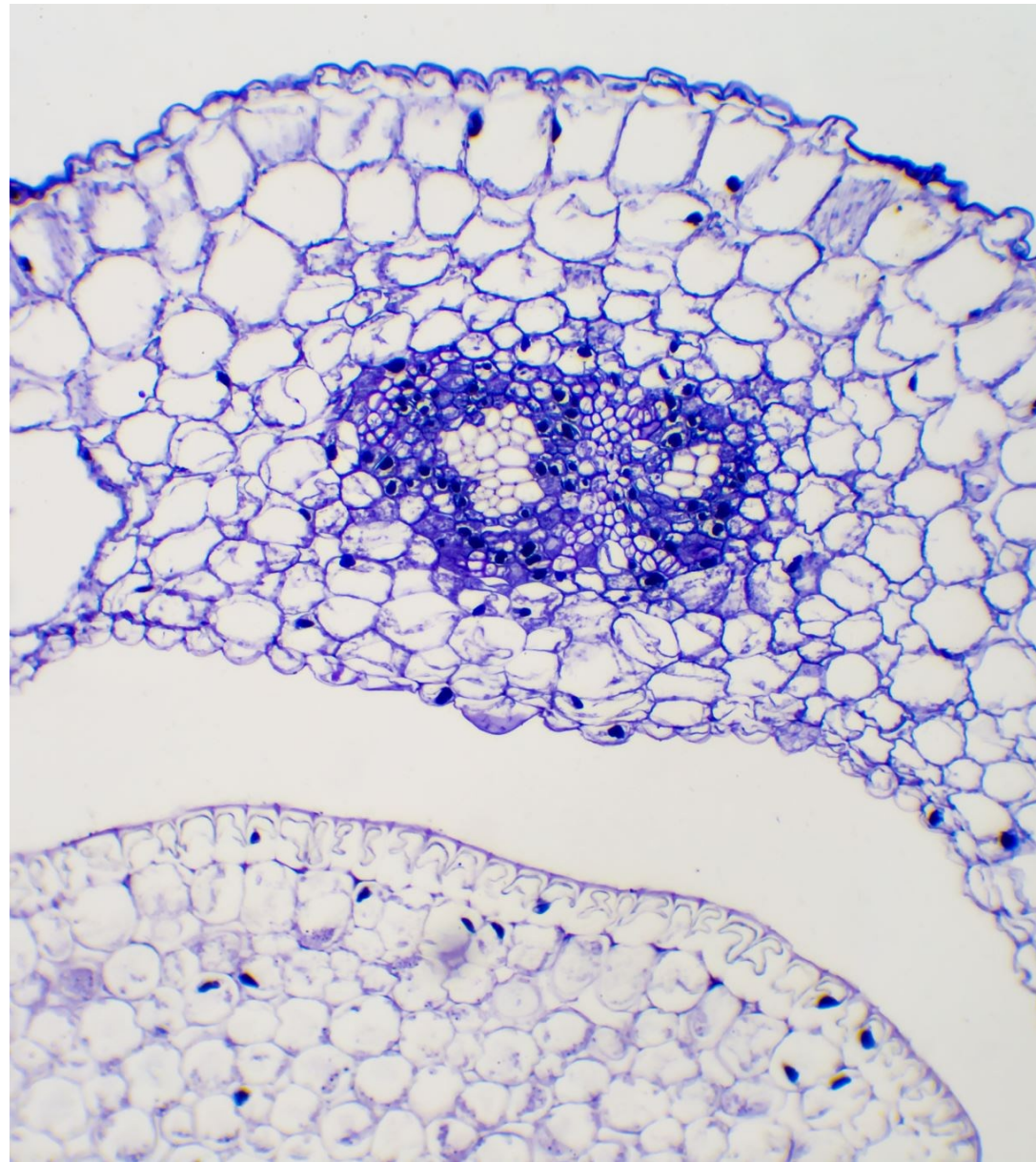
- Catabolic pathways produce energy by **oxidizing** organic fuels
- **Glycolysis** produce chemical energy by oxidizing **glucose to pyruvate**
- After pyruvate is oxidized, **Krebs/citric acid cycle completes** the energy-producing **oxidation** of organic molecules
- During **oxidative phosphorylation**, chemiosmosis **couple electron transport** to **ATP synthesis**
- Fermentation enable cells to produce ATP without oxygen

Why do we need energy?

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“I spent the afternoon digesting my lunch, pumping several quarts of blood through miles of tissue, filtering toxins in my liver, and replacing millions of skin cells. I didn’t have time for anything else.”



Activity

- Define chemiosmosis and explain how a gradient of proton is established across the inner mitochondrial membrane?





Thank You